

Lie Groups: Topology, Connectedness, and Matrix Examples

Road map

Continuous groups contain infinitely many elements that can be arbitrarily close to one another, and their elements are locally described by real parameters. To make these statements precise, this chapter first reviews topological spaces, continuous maps, closure, interior, boundary, compactness, connectedness, simple connectedness, and homeomorphisms. It then introduces topological groups and proves that the identity component is a normal subgroup. Two central matrix examples are treated in detail:

$$O(n) \text{ is compact,} \quad O(n)^\circ = SO(n),$$

and

$$GL(n, \mathbb{R})^\circ = GL^+(n, \mathbb{R}) = \{A \in GL(n, \mathbb{R}) : \det A > 0\}.$$

Finally, we pass from topological groups to Lie groups and linear Lie groups, state the closed subgroup theorem and Ado's theorem in the form needed here, and construct classical, triangular, unipotent, and affine examples.

13.1 Why continuous groups require topology

Unlike a finite group, a continuous group has infinitely many elements and a notion of one element being close to another. For example, a rotation through an angle φ is close to a rotation through $\varphi + \Delta\varphi$ when $\Delta\varphi \rightarrow 0$. Its elements can be parameterized, at least locally, by real numbers. The set of group elements therefore carries extra structure:

- a topology, which makes closeness and continuity meaningful;
- frequently a manifold structure, which permits differentiation.

The main examples in these notes are *linear Lie groups*, realized by matrices. They are especially useful in physics and in quantum mechanics. Historically, some early physicists jokingly called the rapid entrance of group theory into quantum mechanics the *Gruppenpest*, or “group plague.” The matrix approach used here keeps the differential-geometric prerequisites as light as possible without hiding the topology that is actually needed.

Definition 13.1: Topological group

A group G , equipped with a topology, is a *topological group* if the maps

$$m : G \times G \longrightarrow G, \quad (g_1, g_2) \longmapsto g_1 g_2,$$

and

$$\iota : G \longrightarrow G, \quad g \longmapsto g^{-1},$$

are continuous. The domain $G \times G$ carries the product topology.

13.2 Topological spaces and continuous maps

13.2.1 Open sets and the axioms of a topology

Definition 13.2: Topological space

Let X be a set. A *topology* on X is a collection $\mathcal{T} \subseteq \mathcal{P}(X)$ of subsets, whose members are called *open sets*, such that:

1. $\emptyset, X \in \mathcal{T}$;
2. if $\{U_j\}_{j \in J} \subseteq \mathcal{T}$, where J may be infinite, then

$$\bigcup_{j \in J} U_j \in \mathcal{T};$$

3. if $U_1, \dots, U_r \in \mathcal{T}$, where r is finite, then

$$\bigcap_{j=1}^r U_j \in \mathcal{T}.$$

The pair $(X, \mathcal{T})$ is a *topological space*.

The asymmetry between arbitrary unions and only finite intersections is essential. For example,

$$\bigcap_{n=1}^{\infty} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\},$$

which is not open in the usual topology of $\mathbb{R}$.

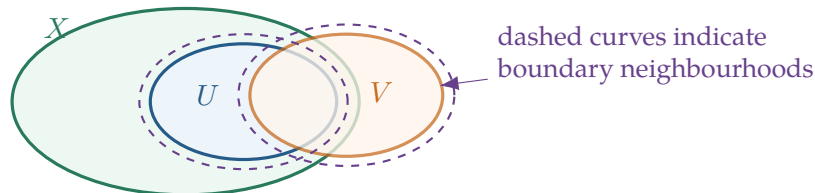


图 13.1: Open sets contain none of their boundary merely by definition; each of their points has a neighbourhood remaining inside the set.

Example 13.1: Metric topology

Let (X, d) be a metric space. For $x \in X$ and $\varepsilon > 0$, the open ball is

$$B_\varepsilon(x) = \{y \in X : d(x, y) < \varepsilon\}.$$

Declare a set open if it is a union of open balls. Arbitrary unions and finite intersections of such sets are again open, so this defines the metric topology. Euclidean spaces $\mathbb{R}^n$ and $\mathbb{C}^n$ always carry this standard topology unless stated otherwise.

Example 13.2: Two extreme topologies

On any set X ,

$$\mathcal{T}_{\text{disc}} = \mathcal{P}(X)$$

is the *discrete topology*: every subset is open. At the opposite extreme,

$$\mathcal{T}_{\text{ind}} = \{\emptyset, X\}$$

is the *indiscrete*, or trivial, topology.

13.2.2 Continuity as preservation of open preimages**Definition 13.3: Continuous map**

Let X and Y be topological spaces. A map $f : X \rightarrow Y$ is *continuous* if

$$U \subseteq Y \text{ open} \implies f^{-1}(U) \subseteq X \text{ open}.$$

The definition concerns inverse images, not images. In fact, every map $f : X \rightarrow Y$ induces on X the collection of inverse images

$$\{f^{-1}(U) : U \subseteq Y \text{ open}\};$$

requiring these sets to be open is exactly what continuity means.

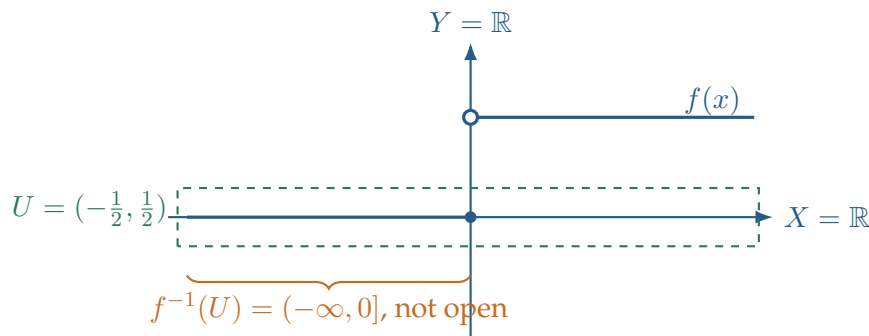


图 13.2: The step function $f(x) = 0$ for $x \leq 0$ and $f(x) = 1$ for $x > 0$ is not continuous: the open set $U = (-\frac{1}{2}, \frac{1}{2})$ has a non-open inverse image.

13.3 Closed sets, closure, interior, and boundary

Definition 13.4: Closed set

A subset $A \subseteq X$ is *closed* if its complement is open:

$$X \setminus A \in \mathcal{T}.$$

Definition 13.5: Closure

The *closure* $\bar{A}$ is the smallest closed subset of X containing A . Equivalently,

$$\bar{A} = \bigcap \{F \subseteq X : F \text{ closed and } A \subseteq F\}.$$

Writing every closed set as $X \setminus U$, this is the formula used in the lecture:

$$\bar{A} = \bigcap \{X \setminus U : U \in \mathcal{T}, A \subseteq X \setminus U\}.$$

Definition 13.6: Interior and boundary

The *interior* of A is the largest open set contained in A :

$$A^\circ = \bigcup \{U \in \mathcal{T} : U \subseteq A\}.$$

The *boundary* of A is

$$\partial A = \bar{A} \setminus A^\circ.$$

These definitions immediately imply the identities recorded in the notes:

$$A \text{ open} \implies A \cap \partial A = \emptyset,$$

$$A \text{ closed} \implies \partial A \subseteq A,$$

$$A^\circ \cap \partial A = \emptyset,$$

$$\bar{A} \cap \partial A = \partial A,$$

$$A^\circ \cup \partial A = \bar{A},$$

$$\bar{A} \cup \partial A = \bar{A}.$$

For $a < b$, the four intervals

$$[a, b], \quad [a, b), \quad (a, b], \quad (a, b)$$

all have the same closure, interior, and boundary in $\mathbb{R}$:

$$\bar{A} = [a, b], \quad A^\circ = (a, b), \quad \partial A = \{a, b\}.$$

13.4 Covers and compactness

Definition 13.7: Cover and open cover

A family $\{A_i\}_{i \in I}$ of subsets of X is a *cover* of X if

$$\bigcup_{i \in I} A_i = X.$$

If every A_i is open, it is an *open cover*.

Definition 13.8: Compact space

A topological space X is *compact* if every open cover $\{U_i\}_{i \in I}$ contains a finite subcover: there exist $i_1, \dots, i_r \in I$ such that

$$X = U_{i_1} \cup \dots \cup U_{i_r}.$$

Theorem 13.1: Heine–Borel theorem

A subset $X \subseteq \mathbb{R}^n$, with the subspace topology, is compact if and only if it is closed and bounded.

The lecture used this theorem without proof. Some immediate examples are:

- $\mathbb{R}_{>0} = \{x \in \mathbb{R} : x > 0\}$ is not compact; it is neither closed in $\mathbb{R}$ nor bounded.
- $[0, \pi] \cup [2, 3]$ is closed and bounded, hence compact. Since $2 < 3 < \pi$, this particular union actually equals $[0, \pi]$.
- The sphere

$$S^n = \{x \in \mathbb{R}^{n+1} : \|x\| = 1\}$$

is closed and bounded, hence compact.

13.5 Connectedness and homeomorphisms

13.5.1 Connected and simply connected spaces

Definition 13.9: Connected space

A topological space X is *connected* if it cannot be written as

$$X = X_1 \cup X_2$$

with X_1 and X_2 nonempty, disjoint, open subsets of X . Such a decomposition, if it exists, is a *separation*; then X is *disconnected*.

Definition 13.10: Loop and simple connectedness

A *loop* in X based at x_0 is a continuous map

$$\gamma : [0, 1] \longrightarrow X, \quad \gamma(0) = \gamma(1) = x_0.$$

A path-connected space is *simply connected* if every loop can be continuously contracted to a point. Explicitly, there must be a continuous map $H : [0, 1] \times [0, 1] \rightarrow X$ such that

$$H(0, t) = \gamma(t), \quad H(1, t) = x_0, \quad H(s, 0) = H(s, 1) = x_0.$$

The n -torus

$$T^n = S^1 \times \cdots \times S^1$$

is connected but not simply connected. Already on T^2 , a loop winding once around the central hole cannot shrink to a point without leaving the surface.

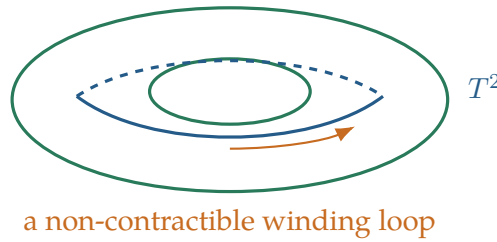


图 13.3: The torus is connected, but the displayed loop cannot be contracted to a point on the torus.

13.5.2 Homeomorphisms

Definition 13.11: Homeomorphism

Let X_1 and X_2 be topological spaces. A map $f : X_1 \rightarrow X_2$ is a *homeomorphism* if it is bijective, continuous, and its inverse $f^{-1} : X_2 \rightarrow X_1$ is continuous. If such a map exists, we write

$$X_1 \cong X_2 \quad \text{or} \quad X_1 \sim X_2$$

and say that the spaces are homeomorphic.

Being homeomorphic is an equivalence relation. The identity gives reflexivity, inverse maps give symmetry, and composition gives transitivity.

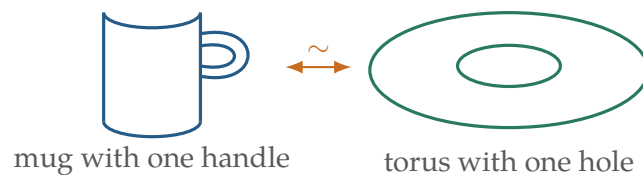


图 13.4: The familiar mug–torus correspondence: topology remembers the number of holes, not lengths or angles.

Similarly, two disjoint circles are homeomorphic as an abstract topological space to two linked circles. Linking is a property of a particular embedding in $\mathbb{R}^3$, not of the circles as a space by themselves. This illustrates why topology permits continuous bending and stretching but not cutting or gluing.

13.6 Topological groups and basic examples

13.6.1 Subgroups and standard examples

Proposition 13.1: Subgroups inherit a topological group structure

If G is a topological group and $H \leq G$ has the subspace topology, then H is a topological group.

Proof. The multiplication on H is the restriction

$$m|_{H \times H} : H \times H \rightarrow H,$$

and inversion is the restriction $\iota|_H : H \rightarrow H$. Restrictions of continuous maps are continuous. ■

Every abstract group becomes a topological group when given the discrete topology. Important nondiscrete examples include:

1. $(\mathbb{R}^n, +)$ with its standard topology.
2. The subgroup $(\mathbb{Z}^n, +) \leq (\mathbb{R}^n, +)$, whose subspace topology is discrete.
3. $(\mathbb{C}, +)$ and $(\mathbb{C}^\times, \cdot)$, where $\mathbb{C}^\times = \mathbb{C} \setminus \{0\}$.

4.

$$S^1 = \{z \in \mathbb{C} : |z| = 1\} \leq \mathbb{C}^\times.$$

5. $(\mathbb{R}^\times, \cdot)$ and its connected subgroup $(\mathbb{R}_{>0}, \cdot)$.

6.

$$GL(n, \mathbb{R}) = \{A \in M_n(\mathbb{R}) : \det A \neq 0\}.$$

Matrix multiplication is polynomial in the entries, and

$$A^{-1} = \frac{\text{adj}(A)}{\det A}$$

is continuous wherever $\det A \neq 0$.

7. The real matrix subgroups $SL(n, \mathbb{R})$, $O(n)$, $SO(n)$, and $Sp(2n, \mathbb{R})$.
8. Their complex analogues $GL(n, \mathbb{C})$, $SL(n, \mathbb{C})$, $U(n)$, and $SU(n)$.

13.6.2 Homomorphisms of topological groups

Definition 13.12: Topological-group homomorphism and isomorphism

For topological groups G and H , a map $f : G \rightarrow H$ is a *homomorphism of topological groups* if it is both a group homomorphism and continuous. It is an *isomorphism of topological groups* if it is a group isomorphism and both f and f^{-1} are continuous.

The examples in the lecture are as follows.

1. If $H \leq G$, the inclusion $H \hookrightarrow G$ is a continuous homomorphism.

2. The exponential map

$$\exp : (\mathbb{R}, +) \longrightarrow (S^1, \cdot), \quad t \longmapsto e^{it},$$

is a continuous surjective homomorphism with

$$\ker(\exp) = 2\pi\mathbb{Z},$$

a discrete topological subgroup of $\mathbb{R}$.

3.

$$\exp : (\mathbb{R}, +) \longrightarrow (\mathbb{R}_{>0}, \cdot), \quad t \longmapsto e^t,$$

is a topological-group isomorphism; its inverse is $\log$.

4. The subgroup

$$H = \left\{ \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix} : t \in \mathbb{R} \right\} \leq GL(2, \mathbb{R})$$

is topologically isomorphic to $(\mathbb{R}, +)$ through

$$t \longmapsto \begin{pmatrix} 1 & t \\ 0 & 1 \end{pmatrix},$$

because multiplication adds the upper-right entries.

5. The determinant

$$\det : GL(n, \mathbb{R}) \longrightarrow \mathbb{R}^\times$$

is a continuous group homomorphism.

13.7 The identity component

Definition 13.13: Identity component

Let G be a topological group with identity e . The *path-connected component of the identity*, denoted G° , is

$$G^\circ = \left\{ g \in G : \begin{array}{l} \text{there is a continuous path } c : [0, 1] \rightarrow G \\ \text{with } c(0) = e \text{ and } c(1) = g \end{array} \right\}.$$

In the finite-dimensional Lie groups considered below, this agrees with the usual connected component of the identity.

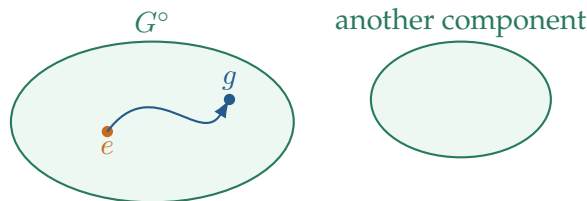


图 13.5: The identity component consists of the elements reachable from e by continuous paths.

Theorem 13.2: Normality of the identity component

For every topological group G , the path component G° is a normal subgroup of G .

Proof. We verify the subgroup properties and normality.

First take $h \in G^\circ$. Choose a path c with $c(0) = e$ and $c(1) = h$. Define

$$\bar{c}(t) = c(1-t)h^{-1}.$$

This is continuous because group multiplication and inversion are continuous. Its endpoints are

$$\bar{c}(0) = c(1)h^{-1} = e, \quad \bar{c}(1) = c(0)h^{-1} = h^{-1}.$$

Hence $h^{-1} \in G^\circ$.

Next let $h_1, h_2 \in G^\circ$, with paths c_1, c_2 from e to h_1, h_2 . Pointwise multiplication gives a continuous path

$$c(t) = c_1(t)c_2(t),$$

and

$$c(0) = ee = e, \quad c(1) = h_1h_2.$$

Thus $h_1h_2 \in G^\circ$, so $G^\circ \leq G$.

Finally, let $g \in G$ and $h \in G^\circ$. If c joins e to h , then

$$c^g(t) = gc(t)g^{-1}$$

is continuous and satisfies

$$c^g(0) = e, \quad c^g(1) = ghg^{-1}.$$

Therefore $ghg^{-1} \in G^\circ$, proving $G^\circ \trianglelefteq G$. ■

13.8 The topology of the orthogonal group

Theorem 13.3: Compactness and identity component of $O(n)$

$$O(n) \text{ is compact,} \quad O(n)^\circ = SO(n).$$

In particular, $SO(n) \trianglelefteq O(n)$.

Compactness

Give $M_n(\mathbb{R}) \cong \mathbb{R}^{n^2}$ the Frobenius norm

$$\|A\|_F^2 = \text{tr}(AA^T) = \sum_{i,j} A_{ij}^2.$$

If $A \in O(n)$, then $AA^T = I$, so

$$\|A\|_F^2 = \text{tr}(I) = n.$$

Hence $O(n)$ is bounded.

The map

$$F : M_n(\mathbb{R}) \longrightarrow M_n(\mathbb{R}), \quad F(A) = AA^T,$$

is continuous because its entries are polynomials in the entries of A . Moreover,

$$O(n) = F^{-1}(\{I\}).$$

The singleton $\{I\}$ is closed, and a continuous inverse image of a closed set is closed. Thus $O(n)$ is closed in $\mathbb{R}^{n^2}$. By the Heine–Borel theorem, it is compact.

$SO(n)$ lies in the identity component

Every real orthogonal matrix is orthogonally conjugate to a block diagonal matrix made of 2×2 rotation blocks

$$R(\alpha) = \begin{pmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{pmatrix}$$

and possible 1×1 blocks (1) or (-1) . When $\det A = 1$, the (-1) -blocks occur in pairs and may be regarded as $R(\pi)$. Consequently, for $A \in SO(n)$ there is an orthogonal P such that

$$P^{-1}AP = \text{diag}(R(\alpha_1), \dots, R(\alpha_r), 1, \dots, 1).$$

The path

$$c(t) = P \text{diag}(R(t\alpha_1), \dots, R(t\alpha_r), 1, \dots, 1)P^{-1}$$

lies in $SO(n)$ and satisfies $c(0) = I$, $c(1) = A$. Hence $SO(n)$ is path connected and

$$SO(n) \subseteq O(n)^\circ.$$

No determinant- -1 matrix lies in the identity component

Suppose that a path $c : [0, 1] \rightarrow O(n)$ joined I to a matrix B with $\det B = -1$. The continuous function

$$t \mapsto \det c(t)$$

would start at $+1$ and end at -1 . The intermediate value theorem would force $\det c(t_0) = 0$ for some t_0 , impossible for an orthogonal matrix. Thus

$$O(n)^\circ \subseteq SO(n).$$

Together with the opposite inclusion, this proves

$$O(n)^\circ = SO(n).$$

Normality also follows either from the identity-component theorem or from

$$SO(n) = \ker(\det : O(n) \rightarrow \{\pm 1\}).$$

13.9 The two components of $GL(n, \mathbb{R})$

Define

$$GL^+(n, \mathbb{R}) = \{A : \det A > 0\}, \quad GL^-(n, \mathbb{R}) = \{A : \det A < 0\}.$$

Theorem 13.4: Identity component of the real general linear group

$$GL(n, \mathbb{R})^\circ = GL^+(n, \mathbb{R}).$$

Thus $GL(n, \mathbb{R})$ has precisely two connected components, distinguished by the sign of the determinant.

Proof. Let $A \in GL^+(n, \mathbb{R})$. Gram–Schmidt, equivalently QR decomposition, produces a unique factorization

$$A = QR,$$

where $Q \in SO(n)$ and R is upper triangular with positive diagonal entries. Write

$$R = DN,$$

where

$$D = \text{diag}(\lambda_1, \dots, \lambda_n), \quad \lambda_i > 0,$$

and N is upper triangular with all diagonal entries equal to 1. Thus

$$A = QDN. \tag{13.1}$$

This is the elementary KDN , or Iwasawa-type, decomposition used in the lecture.

Each factor is path connected to I :

- $Q \in SO(n)$, and the preceding theorem supplies a path $Q(t)$ from I to Q .

-

$$D(t) = \text{diag}(1 - t + t\lambda_1, \dots, 1 - t + t\lambda_n)$$

has strictly positive diagonal entries for all $t \in [0, 1]$, so it joins I to D inside $GL^+(n, \mathbb{R})$.

- Write $N = I + U$, where U is strictly upper triangular. Then

$$N(t) = I + tU$$

remains upper triangular with diagonal entries 1, so it is invertible and joins I to N .

The product

$$c(t) = Q(t)D(t)N(t)$$

is a path in $GL^+(n, \mathbb{R})$ from I to A . Therefore

$$GL^+(n, \mathbb{R}) \subseteq GL(n, \mathbb{R})^\circ.$$

Conversely, no matrix in $GL^-(n, \mathbb{R})$ can be connected to I : the same intermediate-value argument would make the determinant vanish somewhere along the path. Hence

$$GL(n, \mathbb{R})^\circ = GL^+(n, \mathbb{R}).$$

Multiplication by $\text{diag}(-1, 1, \dots, 1)$ is a homeomorphism from $GL^+(n, \mathbb{R})$ to $GL^-(n, \mathbb{R})$, so the negative-determinant part is the other connected component. ■

13.10 Lie groups and linear Lie groups

13.10.1 General definition

Definition 13.14: Lie group

A finite-dimensional *Lie group* is a set G such that:

1. G is a group;
2. G is a finite-dimensional smooth manifold;
3. multiplication $G \times G \rightarrow G$ and inversion $G \rightarrow G$ are smooth.

Smooth maps are continuous, so every Lie group is automatically a topological group. Its dimension is its dimension as a manifold. The hierarchy emphasized in the notes is:

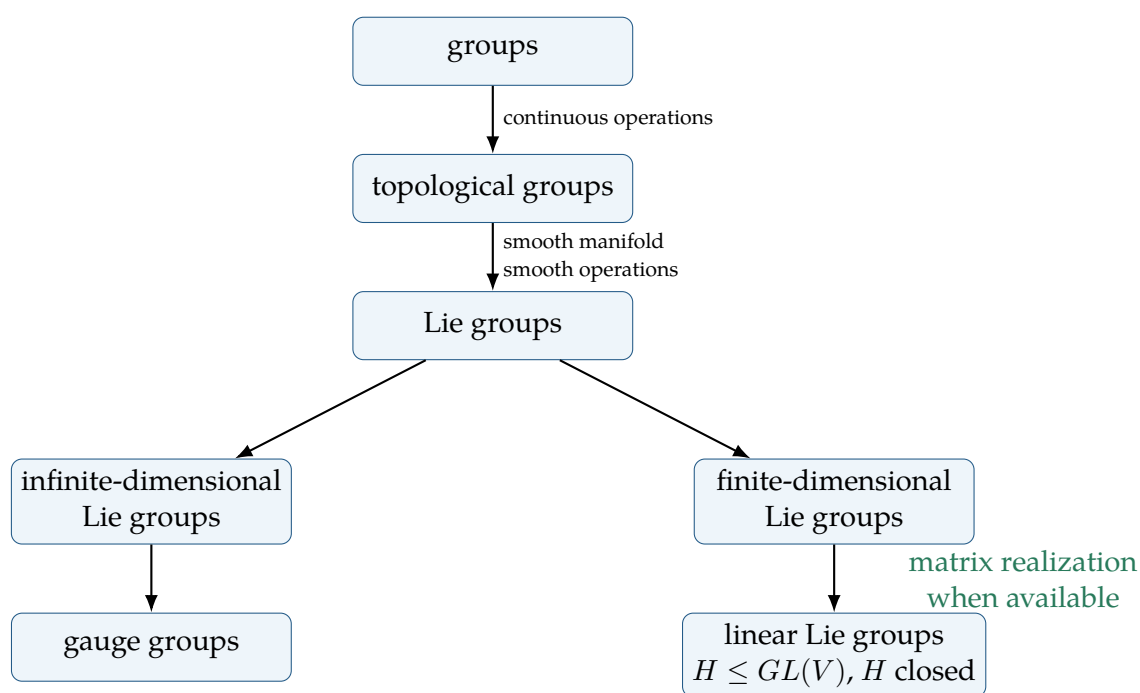


图 13.6: From abstract groups to the matrix Lie groups used in the present course.

Infinite-dimensional gauge groups such as $C^\infty(M, G)$ are important exceptions to the finite-dimensional framework and require additional functional analysis.

13.10.2 Linear Lie groups and the closed subgroup theorem

Definition 13.15: Linear Lie group

For $\mathbb{K} = \mathbb{R}$ or $\mathbb{C}$, a *linear Lie group* is a closed subgroup of $GL(n, \mathbb{K})$ for some n .

Theorem 13.5: Closed subgroup theorem, matrix form

Every closed subgroup H of a finite-dimensional Lie group G is an embedded Lie subgroup. Conversely, an embedded subgroup carrying the subspace topology is closed precisely

when it is a closed embedded Lie subgroup.

In the abbreviated terminology of the handwritten lecture: for a subgroup $H \leq G$, being a matrix Lie subgroup is equivalent to being closed in G . If $H \trianglelefteq G$, then a closed normal subgroup is a normal Lie subgroup.

Theorem 13.6: Ado's theorem, Lie-algebra form

Every finite-dimensional Lie algebra over $\mathbb{R}$ or $\mathbb{C}$ has a faithful finite-dimensional matrix representation.

Ado's theorem explains why matrix methods capture every finite-dimensional Lie algebra. A useful precision is that it is a theorem about Lie *algebras*; a Lie group can still have a global topological obstruction to being faithfully represented by finite matrices. Thus linear Lie groups form the central, but not literally exhaustive, class of finite-dimensional Lie groups.

13.11 Examples of linear Lie groups

13.11.1 Classical groups as automorphism groups of forms

Let $\Phi \in GL(n, \mathbb{K})$ represent a nondegenerate bilinear form. Its automorphism group is

$$\text{Aut}(\Phi) = \{A \in GL(n, \mathbb{K}) : A^T \Phi A = \Phi\}. \quad (13.2)$$

For a sesquilinear form over $\mathbb{C}$, replace transpose by conjugate transpose:

$$\text{Aut}(\Phi) = \{A \in GL(n, \mathbb{C}) : A^\dagger \Phi A = \Phi\}. \quad (13.3)$$

Symmetric, skew-symmetric, and Hermitian choices of Φ produce the orthogonal, symplectic, and unitary families.

These groups are closed. To see this in the bilinear case, suppose $A_r \in \text{Aut}(\Phi)$ and $A_r \rightarrow A \in GL(n, \mathbb{K})$. Then continuity of matrix multiplication and transpose gives

$$A_r^T \Phi A_r = \Phi \implies A^T \Phi A = \lim_{r \rightarrow \infty} A_r^T \Phi A_r = \Phi.$$

Hence $A \in \text{Aut}(\Phi)$. The same argument, with $A^\dagger$, proves the sesquilinear case. Therefore these automorphism groups are linear Lie groups by the closed subgroup theorem.

If $G \leq GL(n, \mathbb{K})$ is a linear Lie group, then

$$H = \{A \in G : \det A = 1\} = (\det|_G)^{-1}(\{1\})$$

is closed because the determinant is continuous. Hence H is again a linear Lie group. This produces, among others, $SO(n)$ from $O(n)$ and $SU(n)$ from $U(n)$.

13.11.2 The Borel and unitriangular groups

The upper triangular, or standard Borel, group is

$$UT_n(\mathbb{K}) = \{A \in GL(n, \mathbb{K}) : A_{ij} = 0 \text{ whenever } i > j\}.$$

Each equation $A_{ij} = 0$ defines a closed condition, so $UT_n(\mathbb{K})$ is closed in $GL(n, \mathbb{K})$ and is therefore a linear Lie group. Schematically, its matrices have the shape

$$\begin{pmatrix} * & * & \cdots & * \\ 0 & * & \cdots & * \\ \vdots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & * \end{pmatrix}, \quad \text{all diagonal entries nonzero.}$$

The special unitriangular group is

$$SUT_n(\mathbb{K}) = \{A \in UT_n(\mathbb{K}) : A_{ii} = 1 \text{ for every } i\}.$$

Its elements are $I + N$ with N strictly upper triangular. Since $N^n = 0$, every such matrix is unipotent:

$$(A - I)^n = 0.$$

This is the *unipotent group of rank n* mentioned in the notes.

13.11.3 The affine group

The affine group is the closed matrix subgroup

$$\text{Aff}_n(\mathbb{K}) = \left\{ \begin{pmatrix} A & t \\ 0 & 1 \end{pmatrix} : A \in GL(n, \mathbb{K}), t \in \mathbb{K}^n \right\} \leq GL(n+1, \mathbb{K}).$$

Embed $x \in \mathbb{K}^n$ into homogeneous coordinates as $\begin{pmatrix} x \\ 1 \end{pmatrix}$. Then

$$\begin{pmatrix} A & t \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ 1 \end{pmatrix} = \begin{pmatrix} Ax + t \\ 1 \end{pmatrix}.$$

Thus $\text{Aff}_n(\mathbb{K})$ acts on $\mathbb{K}^n$ by affine transformations

$$x \mapsto Ax + t.$$

Multiplication displays the semidirect-product law:

$$(A, t)(B, s) = (AB, As + t),$$

so

$$\text{Aff}_n(\mathbb{K}) \cong GL(n, \mathbb{K}) \ltimes \mathbb{K}^n.$$

What to retain from this chapter

1. A topological group is a group whose multiplication and inversion are continuous.
2. A topology is closed under arbitrary unions and finite intersections. Continuity means that inverse images of open sets are open.
3. Closure, interior, and boundary satisfy

$$\partial A = \overline{A} \setminus A^\circ, \quad \overline{A} = A^\circ \cup \partial A.$$

4. Compactness means that every open cover has a finite subcover; in $\mathbb{R}^n$, compactness is equivalent to being closed and bounded.
5. The identity component G° is always a normal subgroup.

6.

$$O(n) \text{ is compact,} \quad O(n)^\circ = SO(n).$$

7.

$$GL(n, \mathbb{R})^\circ = GL^+(n, \mathbb{R}) = \{A : \det A > 0\}.$$

A *QDN* decomposition gives an explicit path to the identity.

8. A Lie group is simultaneously a smooth manifold and a group with smooth operations. A linear Lie group is a closed subgroup of $GL(n, \mathbb{R})$ or $GL(n, \mathbb{C})$.
9. The classical groups, determinant-one subgroups, and the triangular, unitriangular, and affine groups above are all linear Lie groups.